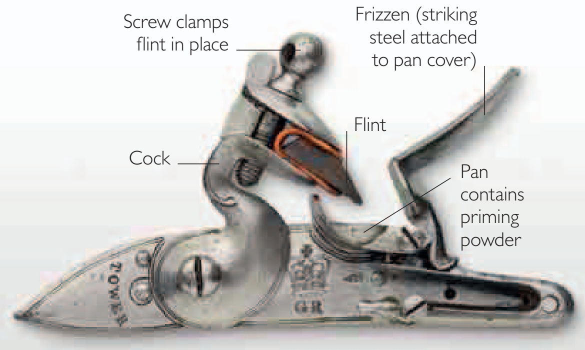
TURNING POINT

GUNS FOR ALL

While the wheel-lock (see pp. 26–27) brought new opportunities for the creation of smaller, more portable firearms, it was a complex design and expensive to build. By the end of the 16th century, efforts to find a reliable but simpler and cheaper mechanism yielded a new lock. This "flintlock" utilized a piece of natural flint to strike hardened steel, generating sparks that ignited the priming powder. Due to their simple, robust working parts, flintlock guns were cheaper and more reliable than earlier arms and became the principal weapons for sporting and military purposes for the next two centuries.



▲ **THE FLINTLOCK MECHANISM**
In this mechanism, the jaws of a spring-loaded cock hold a piece of flint. The cover of the priming pan and a striking steel are united to form a frizzen. A touchhole to the side of the pan connects to the barrel's breech.

The problems faced by users of matchlock weapons (see p. 26) were well-known—wind and rain could extinguish the match-cord or blow exposed priming powder away. As a result, matchlock guns were prone to misfire in bad weather. The smoldering match-cord was also unsafe and inconvenient for the user. An improvement on the matchlock, the wheel-lock, provided an internal system for igniting the priming powder, but it was expensive to manufacture, prone to jam if left spanned (see p. 27) for any length of time, and difficult to maintain in the field. The iron pyrite used in the wheel-lock was soft, and wore out quickly. Soon after the wheel-lock evolved, it became clear that a less costly mechanism for firing a gun was needed. By the 1560s, new gunlocks began to appear. They worked on the principle of striking flint on hardened steel to create sparks.

THE FLINTLOCK MECHANISM

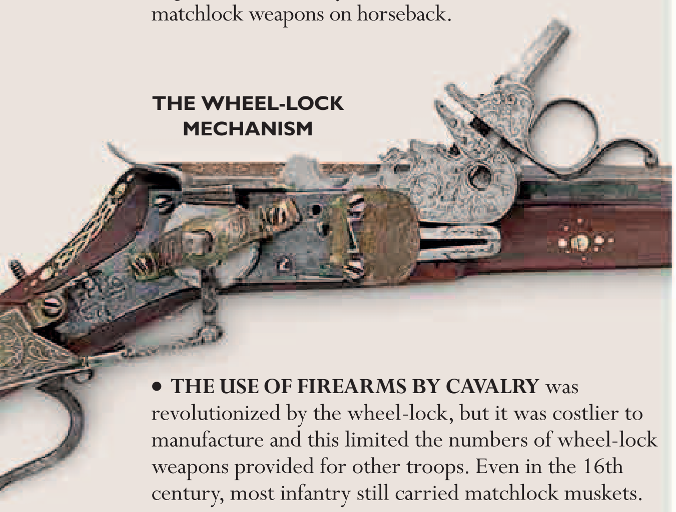
The snaphance, a precursor to the flintlock, was simpler than the wheel-lock. The snaphance's cock held a piece of flint. Pulling the trigger made the cock fall, pushing open the pan cover via an internal link. Simultaneously, the flint scraped against a steel held on a pivoting arm, which produced sparks. These sparks fell into the pan, igniting the priming powder inside. The

» BEFORE

Matchlock and wheel-lock firearms coexisted for a long time, despite the obvious advantages presented by the wheel-lock ignition system. Matchlock weapons were inexpensive and durable and so remained in military service until the latter part of the 17th century.

- **SINGLE-HANDED USE OF FIREARMS** was not possible using the matchlock. It was impractical for cavalry units to load and fire matchlock weapons on horseback.

THE WHEEL-LOCK MECHANISM



- **THE USE OF FIREARMS BY CAVALRY** was revolutionized by the wheel-lock, but it was costlier to manufacture and this limited the numbers of wheel-lock weapons provided for other troops. Even in the 16th century, most infantry still carried matchlock muskets.

- **PORTABLE, HANDHELD GUNS** became a reality in the early 16th century. The wheel-lock enabled guns to be carried primed and ready to fire. As a gun no longer required live fire, it was possible to carry a small weapon in a pocket, spurring the development of the pistol.